

Entropy-Minimizing Control of a Helicopter

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An explicitly parameterized simulation study

Abstract—Native C++ predictive control minimizes physical entropy generation in a simulated electric helicopter with rotor and thermal dynamics. Airborne optimization, PID fallback, and PID landing support are reported separately. Parameters are assumed; a matched PID baseline supports evaluation.

Index Terms—Helicopter, entropy, NMPC.

I. FIX THE VEHICLE AND MISSION

The 2 kg aircraft has main/tail rotor radii 0.55/0.11 m and governed speeds 180/650 rad/s. Its 23 states contain position, velocity, quaternion, body rates, actual pitches, induced velocities, flapping angles, and motor temperatures. Inputs are physical pitch commands:

$$\dot{x} = f(x, u, w), \quad u = [\theta_0, \theta_{1c}, \theta_{1s}, \theta_t]^T. \quad (1)$$

Both controllers share a 70 s takeoff–waypoint–landing mission and exact simulated state/wind. The simulator supplies a spooled trim for each actual mass; controllers do not discover that trim. MuJoCo advances rigid-body/contact dynamics at 0.01 s; rotor and thermal states use RK4. Both controllers share a causal force/velocity mass estimate, assuming inertia scales with mass. The initial 0.1 s trim observation hold is included in mission accounting.

II. CLOSE THE PHYSICAL BALANCES

The boundary includes motor heating, drivetrain loss, aerodynamic dissipation, and wake relaxation at ambient $T_0 = 293.15$ K. For rotor r , near-hover blade-element and inflow relations give [1]

$$P_{a,r} = F_r V_{a,r} + \kappa_r F_r v_{i,r} + P_{\text{profile},r} + \dot{E}_{i,r}. \quad (2)$$

Near-wake storage is $E_{i,r} = M_{i,r} v_{i,r}^2 / 2$. Shaft power includes body rotation. Copper, iron, drivetrain, and braking losses contribute to motor heat L_r . With heat rejection $H_r = h_r(T_r - T_0)$,

$$C_r \dot{T}_r = L_r - H_r, \quad \sigma = \sum_r \left[\frac{L_r}{T_r} + H_r \left(\frac{1}{T_0} - \frac{1}{T_r} \right) \right] + \frac{D_a}{T_0}. \quad (3)$$

D_a comprises modeled wake, profile, translational-drag and angular-damping dissipation. Wind and contact work are separate mechanical ports. Electrical input, storage, heat rejection, and entropy are independently audited.

III. OPTIMIZE THE PHYSICAL OBJECTIVE

Native CasADi/IPOPT [2] seeks local solutions by multiple shooting with 20 intervals of 0.1 s and a 2 s prediction horizon:

$$\min_U J = \int_0^{2s} \sigma(x, u, w) dt + \Phi(x_N), \quad \Phi = \frac{1}{T_0} \sum_r \left[E_{i,r} + C_r \left(T_r - T_0 - T_0 \ln \frac{T_r}{T_0} \right) \right]. \quad (4)$$

Φ accounts for wake relaxation and passive motor cooling. Tracking is constrained, with no tracking-error cost. Bounds cover commands, flight, electrical and thermal states, and terminal settling; they do not prove stability or recursive feasibility. Independently reintegrated feasible solutions drive airborne flight; rejected solves use PID fallback. Before skid contact, geometry triggers explicit PID landing support. This is not contact-optimal control. Playback may be slower than wall time; optimization runs on CPU.

IV. EVIDENCE, LIMITS, AND PROJECT

Five matched 70 s wind/mass cases and a disabled-control test pass, with no audited physical-limit violations. NMPC supplies 90.57–93.29% of all mission decisions; trim observation, fallback, and PID landing remain counted. Committed entropy is 0.12–0.20% lower than PID, exceeding audited accounting and endpoint uncertainty. NMPC tracks less closely within the shared corridor. Four handovers miss the tighter settling target but pass final landing checks. Balance, cooldown, identification, numerical and rejection tests also pass.

The model omits servo electrical losses, rotor-speed transients, stall, vortex-ring flow, and detailed ground aerodynamics. All solves miss the 0.1 s deadline; no hardware efficiency is established. Exergy destruction rate is $T_0 \sigma$ [3]; matched endpoints can equate entropy and dissipated-energy savings. Code, assumed parameters, setup, viewer, and evaluation evidence:

<https://github.com/savethebeesandseeds/Universalis-Materiae-Dexteritas>
Directory: projects/helicopter.

REFERENCES

- [1] W. Johnson, “NDARC: NASA Design and Analysis of Rotorcraft,” NASA, Tech. Rep. TP-2015-218751, 2015, ch. 11.
- [2] CasADi, “Optimal control with CasADi,” Documentation, sec. 8, web.casadi.org/docs.
- [3] F. Philipp *et al.*, “Optimal control of port-Hamiltonian systems: energy, entropy, and exergy,” 2024, arXiv:2306.08914.